Basics.py Code -

```
import cv2
import face_recognition
imgMe = face_recognition.load_image_file('Basics/Me.jpg')
imgMe = cv2.cvtColor(imgMe, cv2.COLOR_BGR2RGB)
imgTest = face_recognition.load_image_file('Basics/yaswanth.jpg')
imgTest = cv2.cvtColor(imgTest, cv2.COLOR_BGR2RGB)
faceLoc = face_recognition.face_locations(imgMe)[0]
encodeMe = face_recognition.face_encodings(imgMe)[0]
cv2.rectangle(imgMe, (faceLoc[3], faceLoc[0]), (faceLoc[1], faceLoc[2]),
(255, 0, 255), 2)
faceLocTest = face_recognition.face_locations(imgTest)[0]
encodeTest = face_recognition.face_encodings(imgTest)[0]
cv2.rectangle(imgTest, (faceLocTest[3], faceLocTest[0]), (faceLocTest[1],
faceLocTest[2]), (255, 0, 255), 2)
results = face_recognition.compare_faces([encodeMe], encodeTest)
faceDis = face_recognition.face_distance([encodeMe], encodeTest)
print(results, faceDis)
cv2.putText(imgTest, f'{results} {round(faceDis[0], 2)}', (50, 50),
cv2.FONT_HERSHEY_COMPLEX, 1, (0, 0, 255), 2)
cv2.imshow('Me', imgMe)
cv2.imshow('Me Test', imgTest)
cv2.waitKey(0)
```

Attendance.py Code-

```
import cv2
import numpy as np
import face_recognition
import os
from datetime import datetime
path = 'ImagesAttendance'
images = []
classNames = []
myList = os.listdir(path)
print(myList)
for cl in myList:
    curImg = cv2.imread(f'{path}/{cl}')
    images.append(curImg)
    classNames.append(os.path.splitext(cl)[0])
print(classNames)
import cv2
import numpy as np
import face_recognition
import os
from datetime import datetime
path = 'ImagesAttendance'
images = []
classNames = []
myList = os.listdir(path)
print(myList)
for cl in myList:
    curImg = cv2.imread(f'{path}/{cl}')
    images.append(curImg)
    classNames.append(os.path.splitext(cl)[0])
print(classNames)
def findEncodings(image):
    encodeList = []
    for img in image:
```

```
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
        encode = face_recognition.face_encodings(img)[0]
        encodeList.append(encode)
    return encodeList
def markAttendance(names):
    with open('Attendance.csv', 'r+') as f:
        myDataList = f.readlines()
        nameList = []
        for line in myDataList:
            entry = line.split(',')
            nameList.append(entry[0])
        if names not in nameList:
            now = datetime.now()
            dtString = now.strftime('%H:%M:%S\n')
            f.writelines(f'\n{names},{dtString}')
encodeListKnown = findEncodings(images)
print('Encoding Complete')
cap = cv2.VideoCapture(0)
while True:
    success, img = cap.read()
    imgS = cv2.resize(img, (0, 0), None, 0.25, 0.25)
    imgS = cv2.cvtColor(imgS, cv2.COLOR_BGR2RGB)
    facesCurFrame = face_recognition.face_locations(imgS)
    encodesCurFrame = face_recognition.face_encodings(imgS,
facesCurFrame)
    for encodeFace, faceLoc in zip(encodesCurFrame, facesCurFrame):
        matches = face_recognition.compare_faces(encodeListKnown,
encodeFace)
        faceDis = face_recognition.face_distance(encodeListKnown,
encodeFace)
        matchIndex = np.argmin(faceDis)
        if matches[matchIndex]:
            name = classNames[matchIndex].upper()
            # print(name)
```

```
        y1, x2, y2, x1 = faceLoc
        y1, x2, y2, x1 = y1 * 4, x2 * 4, y2 * 4, x1 * 4
        cv2.rectangle(img, (x1, y1), (x2, y2), (0, 255, 0), 2)
        cv2.rectangle(img, (x1, y2 - 35), (x2, y2), (0, 255, 0), cv2.FILLED)
        cv2.putText(img, name, (x1 + 6, y2 - 6),
cv2.FONT_HERSHEY_COMPLEX, 1, (255, 255, 255), 2)
        markAttendance(name)
    cv2.imshow('Webcam', img)
    cv2.waitKey(1)
```